FIBRE PRODUCTS IN A FIBRED CATEGORY EXIST WHEN THE BASE FIBRE PRODUCT DOES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 6873-6879, following the statement at 6864-6871.

Editorial note. The proof is a complete instantiation of the cited lemma (carded separately with a full proof); the two remaining conclusion components follow from base-preserving isomorphism invariance of strong cartesian-ness, recorded in the card's obligations.

Allowed standard prerequisites (editorial).

- Fibre product from strongly cartesian lifts. lemma-strongly-cartesian-fibre-product (carded separately), cited at proof line 6878. Supplies that with x -> y strongly cartesian, the base fibre product existing, and a strongly cartesian lift a: w -> z over pr_2 with p(w) = p(x) x p(y) p(z), the fibre product x x_y z exists and is isomorphic to w.
- Existence of strongly cartesian lifts in a fibred category. the fibred category hypothesis of the statement, used at proof lines 6874-6876. Supplies the strongly cartesian morphism pr_2^*z -> z over pr_2 : p(x) x p(y) p(z) -> p(z) used as the object w.

Source reference. Original lemma 06N4: sources/stacks-categories/pinned-original-source.tex, source lines 6531–6531. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $p : \mathcal{S} \to \mathcal{C}$ be a fibred category. Let $x \to y$ and $z \to y$ be morphisms of $\mathcal{S}$ with $x \to y$ strongly cartesian. If $p(x) \times_{p(y)} p(z)$ exists, then $x \times_y z$ exists, $p(x \times_y z) = p(x) \times_{p(y)} p(z)$, and $x \times_y z \to z$ is strongly cartesian.*

Proof. Pick a strongly cartesian morphism $\mathrm{pr}_2^* z \to z$ lying over $\mathrm{pr}_2 : p(x) \times_{p(y)} p(z) \to p(z)$. Then $\mathrm{pr}_2^* z = x \times_y z$ by Lemma 06N4. □

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.